# Department of Information Technology
# Central University of Kashmir

## B.Tech. CSE

| Semester V | | | | | |
|---|---|---|---|---|---|
| S.No. | Course Code | Course Title | Credits | CIA | ESE |
| 1 | BTCS 501 | Computer Organisation & Architecture | 4 | 50 | 50 |
| 2 | BTCS 507 | Computer Graphics | 3 | 50 | 50 |
| 3 | BTCS 508 | Data Communication | 4 | 50 | 50 |
| 4 | BTCS 509 | Operating Systems | 4 | 50 | 50 |
| 5 | BTCS 511 L | Computer Graphics Lab | 2 | 50 | 50 |
| 6 | BTCS 512 | Web design and development | 4 | 50 | 50 |
| 7 | BTCS 513L | Web Design and development lab | 2 | 50 | 50 |
| | | **Total** | **23** | | |

**Central University of Kashmir**

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|---|---|---|---|---|
| **Course Title** | Computer Organization & Architecture | **Course Code** | BTCS 501 | |
| **Degree** | Bachelor of Technology | **Branch** | Computer Science & Engg. | |
| **Course Type** | Theory | **Department** | Information Technology | |
| **Credits** | 04 | **L** | **T** | **P** |
| **Semester** | 5th | **4** | 0 | 0 |

**Course Objectives**

- Define the concept of architecture and incorporate parameters to evaluate and analyze the performance
- Identify the pipelining as a basic technique for increasing CPU performance as well as design, planning and control of pipeline units
- Understanding the evolution of the architectures and the differences between CISC and RISC approaches
- Explain techniques for improving the performance of memory and input/output system
- Recognize the limitations of classical architectures and the importance of parallelism

**Learning Outcomes**

- To apply the knowledge of performance metrics to find the performance of systems.
- To identify high performance architecture design.
- To identify the problems in components of computer.
- To develop independent learning skills and be able to learn more about different computer architectures and hardware.

| Course Outline / Content | | |
|---|---|---|
| **Unit** | **Topics** | **Weeks** |
| 1. | Basics of Computer Architecture and Organization, Stored Program Organization (Von Neumann Architecture), Register Transfer Language, Microoperations(Arithmetic, Logic, Shift), Instruction codes, Registers, Instruction set, Hard wired Control Unit, Instruction cycle, Instruction types: Memory reference, Register Reference and I/O instructions. Interrupt cycle, Computer Organization and Design | 4 |
| 2. | Instruction Formats, Addressing Modes, Stack Organization, Program Control, Characteristics of RISC and CISC, Introduction to Pipelining. Integer and Floating-Point Representation, Fixed Point Arithmetic: Addition, Subtraction, Multiplication and Division With Flowcharts, Floating Point Arithmetic: Addition and Subtraction. | 4 |
| 3. | Input-Output Organization, Peripheral Devices, I/O interface, Isolated and Memory Mapped I/O Asynchronous Data transfer: Strobe control , Handshaking, serial transfer, Priority Interrupt: Daisy chaining, Parallel Priority interrupt, Direct Memory Access | 3 |
| 4. | Memory Hierarchy, Main Memory: RAM, ROM, Auxiliary Memory (Magnetic Disk), Associative Memory, Cache Memory: Mapping Functions, Replacement Algorithm, Virtual Memory Concepts, Virtual Memory Address Translation | 4 |
| **Text Books** | | |
| 1. | Moris Mano, Computer system Architecture, PH | |
| 2. | Hamacher, Computer Organization, McGraw Hill | |
| **References** | | |

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|---|---|
| 1. | Parthasarthy, Advanced Computer Architecture, Cengage India. |
| 2. | Tennenbaum A. S., Structured Computer Organization, PHI. |
| 3. | Gear C. W., Computer Organization and Programming, McGraw Hil |

| Central University of Kashmir | | | | |
|---|---|---|---|---|
| **Course Title** | Computer Graphics | **Course Code** | BTCS 507 | |
| **Degree** | Bachelor of Technology | **Branch** | Computer Science & Engg. | |
| **Course Type** | Theory | **Department** | Information Technology | |
| **Credits** | 03 | **L** | **T** | **P** |
| **Semester** | 5th | **3** | 0 | 0 |

**Course Objectives**

- To understand the basics of various inputs and output computer graphics hardware devices.
- Exploration of fundamental concepts in 2D and 3D computer graphics.
- To know the working of multimedia tools

**Learning Outcomes**

- Students will get the concepts of Graphics display devices, techniques, and different types of graphics drawing algorithms.
- Students will get the concepts of 2D and 3D Geometrical Transformations
- Students will get the concepts of Viewing, Curves and surfaces
- Students will get the concepts of Hidden Line/surface elimination techniques
- Students will get the concepts of some Scan Conversion algorithms.

**Course Synopsis**

Computer Graphicsis a 4 credit course that introduces the concepts and implementation of computer graphics. As one of the important subject areas of the study of computer science and information systems, this course starts with an overview of interactive computer graphics, two dimensional system and mapping, then it presents the most important drawing algorithm, two-dimensional transformation; Clipping, filling, curves and surfaces and an introduction to 3-D graphic

| Course Outline / Content | | |
|---|---|---|
| **Unit** | **Topics** | **Weeks** |
| 1. | Introduction to Computer Graphics: Applications of Computer Graphics. Graphic Display Devices_ Raster and Random. Flat panel display devices, Display Processor, Display Buffer, Concept of Double Buffering and Segmentation of Display Buffer. Color Displays: Shadow Masking and Beam Penetration methods, Use of Lookup tables. Graphics Input and Output Devices_ Description and Applications. Graphic Kernel System, Introduction to GKS, GKS primitives | 4 |
| 2. | 2-D Graphics: Cartesian and Homogeneous Coordinate Systems. Line drawing algorithms (Bresenhams and DDA). Bresenhams Circle and Ellipse Drawing Algorithms. Character generation, 2- Dimensional Transformations. Concepts of Window & Viewport, Window to Viewport Transformations. Filling, Boundary and Floodfill algorithms | 4 |
| 3. | Clipping: Line Clipping Algorithms (Cohen-Sutherland Algorithm), Sutherland Hodgeman Polygon Clipping, Text Clipping, 3-D Graphics, Projections: perspective and parallel projection transformations. 3-Dimensional Transformations. Hidden Surface Removal Techniques, Z-Buffer Algorithm, Back Face Detection. Scan Line Algorithm. Painter's Algorithm. | 4 |
| 4. | Curves and Surfaces: Interpolation, Spline representation, Interpolation & Approximation Splines, Spline Specifications, 04 5 Hermite Interpolation. Beizer-Curves & Surfaces, B-Spline Curves surface | 4 |

| Text Books/References | |
|---|---|
| 1. | Hearn and Baker " Computer Graphics", Pearson Education |

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|---|---|
| 2. | W.M.Newman and Sproull. “Principles of interactive Computer Graphics” ,TMH . |
| 3. | Steven Harrington.” Computer Graphics a Programming Approach” McGraw Hil |
| 4. | James. D. Foley, AVandametal “Computer Graphics” Pearson |
| 5 | David F Frogers and J Alan Adams. “Procedural Elements of Computer Graphics” McGraw Hill |
| 6 | David F Rogers and J Alan Adams. “Mathematical Elements of Computer Graphics” McGraw Hill |

**Central University of Kashmir**

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|---|---|---|---|---|
| **Course Title** | Data Communication | **Course Code** | BTCS 508 | |
| **Degree** | Bachelor of Technology | **Branch** | Computer Science & Engg. | |
| **Course Type** | Theory | **Department** | Information Technology | |
| **Credits** | 04 | **L** | **T** | **P** |
| **Semester** | 5th | **4** | 0 | 0 |

**Course Objectives**

• To understand the concept of communication engineering, signals, channels andcommunication systems.

• To understand and analyze the characteristics of various modulation techniques.

• To critically analyze various modulation techniques used in modern communication systems

• To apply knowledge of computers, software, networking technologies andinformation assurance to an organization's management, operations, and requirements

**Learning Outcomes**

By the end of this course, the student will be able to:

•To Focus on information sharing and networks.

•To Introduce flow of data, categories of network, different topologies.

•To Focus on different coding schemes.

•Brief the students regarding protocols and standards.

•To give clear idea of signals, transmission media, errors in data communications andtheir correction, networks classes and devices,etc.

• Identify various components in a data communication system, describe theirproperties, explain how they work and evaluate their performance

| Course Synopsis | | |
|---|---|---|
| This is a first class on the fundamentals of data communication networks, their architecture, principles of operations, and performance analyses. The goal will be to give some insight into the rationale of why networks are structured the way they are today and to understand the issues facing the designers of next-generation data networks | | |
| **Course Outline / Content** | | |
| **Unit** | **Topics** | **Weeks** |
| 1. | Data and Signals: Data, Signals, Types of Signals, Analog and Digital, Bandwidth,spectrum, Digitization of analog signals, sampling, Nyquistsampling theorem, quantization, quantization noise, Pulse code modulation. Digital modulation Techniques: ASK, FSK, PSK, DPSK, Mary PSK, QAM. Signal constellation. | 4 |
| 2. | Line coding techniques: NRZ, RZ, Biphase, Manchester coding, AMI, HDBn. Transmission media: Guided and un-guided media, twisted wire pair, co-axial cable, optical fibre, microwave links, satellite microwave link, their characteristic features and applications for data transmission, Performance, Wavelength , Shannon Capacity , Media Comparison, PSTN ,Switching | 4 |
| 3. | Data transmission: simplex, half duplex and full duplex, Asynchronous and synchronous data transmission. Carrier, bit and frame synchronization techniques, Phase lock loop. Multiplexing Techniques: Frequency Division Multiplexing, Time Division Multiplexing, Wavelength division Multiplexing and Code Division Multiplexing. Spread Spectrum | 4 |
| 4. | Errors in data communication: Types of errors, error detection and correction techniques, forward error correction, polynomial error detection scheme, computation of CRC. | 4 |

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|---|---|---|
| | Hardware. Study of DTE-DCE in brief:Digital data transmission, DTE-DCE Interface, Modems, 56K Modems ,Cable Modem | |
| **Text Books** | | |
| 1. | Data communication & Networking by BahrouzForouzan | |
| 2. | Sklar, “Digital Communications fundamentals & Applications” Pearson | |
| **References** | | |
| 1. | Computer Networks by Andrew S. Tanenbaum. | |
| 2. | Data and Computer Communications by William Stallings | |
| 3. | Communication Systems by Simon Haykin | |
| 4 | Analog and Digital communication by Sam Shanmugan | |

| Central University of Kashmir | | | | |
|---|---|---|---|---|
| **Course Title** | Operating Systems | **Course Code** | BTCS 509 | |
| **Degree** | Bachelor of Technology | **Branch** | Computer Science & Engg. | |
| **Course Type** | Theory | **Department** | Information Technology | |
| **Credits** | 04 | **L** | **T** | **P** |
| **Semester** | 5th | **4** | 0 | 0 |

**Course Objectives**

To understand the services provided by and the design of an operating system.

- To understand the structure and organization of the file system.
- To understand what a process is and how processes are synchronized and scheduled.
- To understand different approaches to memory management.
- Students should be able to use system calls for managing processes, memory and the file system.
- Students should understand the data structures and algorithms used to implement an OS

**Learning Outcomes**

On completing this course the students should have acquired the following capabilities

- An appreciation of the role of an operating system.
- Become aware of the issues in the management of resources like processor, memory and input-output.
- Should be able to select appropriate productivity enhancing tools or utilities for specific needs like filters or version control.
- Obtain some insight into the design of an operating system.

| **Course Synopsis** | | |
|---|---|---|
| The course will introduce Operating Systems (OS), their design and Implementation. We will discuss the goals of an OS, and some successful and not-so successful OS designs. We will also discuss the following OS services in detail: threadscheduling, security, virtual memory, and file system. In this course we will explore the coreprinciples of operating systems design and implementation, including basic operating systemstructure; process and thread synchronization and concurrency; file systems and storageservers; memory management techniques; process scheduling and resource management; virtualization; and security. | | |
| **Course Outline / Content** | | |
| **Unit** | **Topics** | **Weeks** |
| 1. | Introduction: Computer System Overview-Basic Elements, Instruction Execution, Operating system functions and structure, Evolution of Operating System , Interrupts, Memory Hierarchy, Cache Memory, Direct Memory Access, Multiprocessor and Multicore Organization. Operating system overview-objectives and their functions, Distributed Operating System. | 3 |
| 2. | Process: Process Concept, Process States, Process Description and Process Control, Context switching Interprocess Communication, Processes and Threads, Types of Threads, Multicore and Multithreading, Process Scheduling: Definition , Scheduling objectives ,Types of Schedulers, Scheduling criteria : CPU utilization, Throughput, Turnaround Time, Waiting Time. Scheduling algorithms : Preemptive and Non pre-emptive, FCFS – SJF – RR , Multiprocessor scheduling: Types, Performance evaluation of the scheduling. Interprocess Communication. Principles of Concurrency - Mutual Exclusion, Semaphores, Monitors, Readers/Writers problem. Deadlocks – prevention- avoidance – detection | 4 |

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| 3. | Basic Memory Management: Definition, Requirements, Partitioning, Paging and Segmentation, Virtual Memory, Virtual Memory Management , Locality of Refrence, Page Fault , Hardware and Control Structures, Operating System Software, Linux Memory Management, Windows Memory Management. Internal and External Fragmentation . | 4 |
| 4. | I/O Management and Disk Scheduling : I/O devices, organization of I/O functions; OS Design Issues, I/O Buffering, Disk Scheduling, RAID, Disk cache. File management – File Concept, Organization, Directories, File sharing, Access Methods and Record blocking, Secondary Storage Management. | 4 |
| | **Text Books** | |
| 1. | Silberschatz, Peter Galvin, Greg gagne "Operating System Principles". | |
| 2. | Andrew S. Tannenbaum & Albert S. Woodhull, "Operating System Design and Implementation", Prentice Hall | |
| 3. | William Stallings, "Operating Systems – internals and design principles", Prentice Hal | |
| | **References** | |
| 1. | Andrew S. Tannenbaum, "Modern Operating Systems", Prentice Hall | |
| 2. | Gary J.Nutt, "Operating Systems", Pearson/Addison Wesley. | |
| 3. | Pramod Chandra P.Bhatt, "An Introduction to Operating Systems Concepts and Practice" | |

## Central University of Kashmir

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|---|---|---|---|---|
| **Course Title** | Computer Graphics Lab | **Course Code** | BTCS 511 L | |
| **Degree** | Bachelor of Technology | **Branch** | Computer Science & Engg. | |
| **Course Type** | Theory/ Laboratory | **Department** | Information Technology | |
| **Credits** | 02 | **L** | **T** | **P** |
| **Semester** | 5th | **0** | 0 | 2 |

### Course Objectives

- Understand the need of developing graphics application
- Learn algorithmic development of graphics primitives like: line, circle, polygon etc.
- Learn the representation and transformation of graphical images and pictures.

### Learning Outcomes

- Design and implementation of various algorithms to draw a number of shapes
- Design and implementation of various algorithms for designing animation graphics and composite objects
- Design and simulation of various algorithms using multimedia tools

### Course Synopsis

This course provides a broad overview of the basic concepts of computer graphics. Both 2d raster graphics and 3d graphics will be covered. The aim of this course isto provide hands-on exposure to tools, techniques and algorithms in computer graphics

### Course Outline / Content

| **Lab** | **Weeks** |
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|---|---|---|
| | 1. To draw a line using DDA Algorithm.<br>2. To draw a line using Bresenham’s Algorithm.<br>3. To draw a circle using trigonometric Algorithm.<br>4. To draw a circle using Bresenham’s Algorithm.<br>5. To implement polygon boundary fill algorithm.<br>6. To implement polygon flood fill algorithm.<br>7. To translate an object with translation parameters in X and Y directions.<br>8. To scale an object with scaling factors along X and Y directions.<br>9. To rotate an object with a certain angle.<br>10. To perform composite transformations of an object.<br>11. Implementation of simple graphics animation | 10 |
| | **Text Books** | |
| 1. | Hearn and Baker “ Computer Graphics”, Pearson Education | |
| 2. | W.M.Newman and Sproull. “Principles of interactive Computer Graphics” ,TMH | |
| 3. | Steven Harrington.” Computer Graphics a Programming Approach” McGraw Hil | |
| 4. | James. D. Foley, AVandametal “Computer Graphics” Pearson | |
| 5. | David F Frogers and J Alan Adams. “Procedural Elements of Computer Graphics” McGraw Hill | |
| 6. | . David F Rogers and J Alan Adams. “Mathematical Elements of Computer Graphics” McGraw Hill | |
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<table>
<tr><td colspan="5">Central University of Kashmir</td></tr>
<tr><td>Course Title</td><td>Web Design & Development</td><td>Course Code</td><td colspan="2">BTCS 512</td></tr>
<tr><td>Degree</td><td>Bachelor of Technology</td><td>Branch</td><td colspan="2">Computer Science & Engg.</td></tr>
<tr><td>Course Type</td><td>Theory</td><td>Department</td><td colspan="2">Information Technology</td></tr>
<tr><td>Credits</td><td>04</td><td>L</td><td>T</td><td>P</td></tr>
<tr><td>Semester</td><td>5th</td><td>4</td><td>0</td><td>0</td></tr>
<tr><td colspan="5">Course Objectives</td></tr>
<tr><td colspan="5">• To understand the basics of HTML syntax, including the head and body sections, creating hyperlinks, table layout, and presentation, and forms, as well as advanced concepts such as using frames and working with images.<br>• To understand cascading style sheets (CSS), including positioning elements using static, relative, and absolute positioning, writing CSS selectors and rules, and understanding the cascade, inheritance, and specificity.<br>• To understand the basics of HTML forms, including different input types such as textboxes, radio buttons, and checkboxes, as well as how to use JavaScript for form validation and event handling.<br>• To understand JavaScript objects and data types, the Document Object Model, and Dynamic HTML using JavaScript and CSS for advanced form validation, as well as an overview of site redesign.<br>• To understand form handling, form data processing, query strings, cookies and sessions.<br>• To understand the basic syntax and commands of PHP, how to connect PHP to a MySQL server and perform various database operations such as creating databases and tables, inserting data, executing queries, and deleting data and tables</td></tr>
<tr><td colspan="5">Learning Outcomes</td></tr>
</table>

- Apply the Knowledge to create dynamic and interactive web applications that respond to user input and events.
- Apply the knowledge of PHP, to create dynamic web pages by incorporating serverside scripting to interact with databases, process form data, and handle user authentication through the use of cookies and sessions.
- To apply the knowledge to create various web projects ranging from simple personal websites to complex e-commerce sites, web applications, and more.

**Course Synopsis**

Student will understand the various aspects of web development, including internet and web architecture, client-side and server-side web programming, HTML and CSS syntax, form elements, JavaScript programming, the Document Object Model (DOM), PHP syntax, and connecting to and using MySQL databases. Students can apply this knowledge to create websites, build dynamic web applications, and perform database operations on web servers

**Course Outline / Content**

| Unit | Topics | Weeks |
|---|---|---|
| 1. | **HTML & Introduction to CSS:**<br>Internet and World Wide Web, Architecture, Role of Webservers, DNS, Web Browsers, Web Programming; Client side Vs Server side. HTML – Concepts of Hypertext, Elements of HTML syntax ,Head and Body Sections, Building HTML documents, Inserting texts, Images, Hyperlinks, Backgrounds and Color controls, Table layout and presentation, List types and its tags, HTML Forms and Form Elements.HTML rules: Extensible markup languages; Frames: frame and frameset properties. Images: Image types (JPG, GIF, PNG). Inline, embedded, and external styles. Writing Style Rules: Writing CSS selectors and rules to tie style attributes and values to html elements. The cascade: | 4 |

| | | |
|---|---|---|
| | Inheritance, specificity, and the cascade.CSS positioning: Static, relative, and absolute positioning | |
| 2. | **Advanced CSS, and Java Script programming:**<br><br>HTML Form Basics,The form element and inputs: textbox, radio buttons, checkbox, textarea.Java Script (Constants, Variables and data types, Operators, Window Object, document object and string function). JavaScript Objects (Math, String, Array, Date), Event Handling and Form Validation.The DOM and JavaScript Object Models: The W3C Document Object Model; using nodes; DHTML: JavaScript + CSS = Dynamic HTML, Advanced form validation: JavaScript's innerHTML and dynamic CSS for advanced form validation. Advanced JavaScript— Super Hypertexts: Finding. Web Site Design / Redesign: Overview of site redesign. | 4 |
| 3. | **PHP:**<br><br>Basic Syntax of PHP, Variables and constants, Data type, Operators, Decisions, Lopping, Arrays, Strings, Functions, superglobals, form handling, Form Data Processing, querystrings, Cookies and sessions, PHP and XML | 3 |
| 4. | **PHP and MySQL:**<br><br>Basic commands with PHP examples, Connection to sever-Connect to MYSQL from PHP. Understanding Database and structure, creating database, selecting a database, listing database, listing table names, creating a table, inserting data, queries, deleting database, deleting data and tables, Understanding phpMyAdmin and features, web hosting, SQL injection. | 4 |
| | **Text Books** | |
| 1. | Headfirst PHP & MySQL, O'REILLY Media. | |

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|---|---|
| 2. | Raj Kamal, Internet & Web Design, Tata McGraw Hill. |
| 3. | Mastering HTML, CSS & JavaScript Web Publishing, Laura Lemay. |
| 4. | PHP and MySQL Web Development, Fifth edition. |
| **References** | |
| 1. | Jon Duckett, HTML & CSS Design and Build Websites, JOHN WILEY & SONS, INC |
| 2. | Thomas A. Powell, HTML & CSS: The Complete Reference, Fifth edition |
| 3. | Robin Nixon, Learning PHP, MYSQL, JavaScript, CSS & HTML5, O'REILLY Media, Inc. |
| 4. | Steven Holzner, PHP: The Complete Reference |

<table>
<tr><td colspan="5">Central University of Kashmir</td></tr>
<tr><td>Course Title</td><td>Web Design & Development Lab</td><td>Course Code</td><td colspan="2">BTCS 513 L</td></tr>
<tr><td>Degree</td><td>Bachelor of Technology</td><td>Branch</td><td colspan="2">Computer Science & Engg.</td></tr>
<tr><td>Course Type</td><td>Laboratory</td><td>Department</td><td colspan="2">Information Technology</td></tr>
<tr><td>Credits</td><td>02</td><td>L</td><td>T</td><td>P</td></tr>
<tr><td>Semester</td><td>5th</td><td>0</td><td>0</td><td>2</td></tr>
<tr><td colspan="5">Course Objectives</td></tr>
<tr><td colspan="5">• Apply the knowledge to manage and to handle web site design and development to solve the real world problems.<br>• Gain a reputed designation as good web designer and web developer ethically by applying advance web technologies<br>• By learning java script able to handle events and client-side processing.<br>• Able to apply the basics to design a web page and identify its elements and attributes.</td></tr>
<tr><td colspan="5">Learning Outcomes</td></tr>
<tr><td colspan="5">Upon successful completion of the course, students should be able to:<br>• Demonstrate an understanding of Web Development Lab<br>• They are able to understand the difference among the various scripting languages.<br>• Learning the basics about web browsing by HTTP and URL<br>• Use JavaScript to create functional forms.<br>• Demonstrate proficiency in using web site design and development according to market demand.<br>• Can show their ability to apply conceptual skills of Web Site Design and Development</td></tr>
</table>

| Course Synopsis | |
|---|---|
| • Develop and demonstrate a HTML file that includes:<br>-HTML element<br>- inserting images, hyperlinks and anchor tags<br>-form and form elements<br>-table layout<br>-list types<br>• Develop and demonstrate, using CSS, a HTML document that uses inline, internal and external CSS.<br>• Develop and demonstrate, using JavaScript, a HTML document in which an Event handler must be included for the form element that collects this information to validate the input. Messages in the alert windows must be produced when errors are detected.<br>• Developing programs that deal file handling in PHP.<br>• Developing programs to understand the concept of Sessions and Cookies in PHP.<br>• Developing programs to understand database connectivity. | |
| **Course Outline / Content** | |
| **Lab** | **Weeks** |

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|---|---|---|
| | 1. Design a registration form in HTML, design home page i.e. educational details, achievements, hobbies and create hyperlinks in home page.<br>2. Design a table that will display student's information .<br>3. Write an HTML code to develop a Web page having two frames that 90 divide the Web page into two equal rows and then divide the second row into equal columns.<br>4. Use of lists and forms in webpages.<br>5. Design a webpage using CSS (inline, internal, & external CSS).<br>6. Embedding JavaScript in HTML pages.<br>7. Design a webpage whose content can be changed using JavaScript like events.<br>8. Validate the registration, user login, user profile and payment by credit card pages using JavaScript.<br>9. Write a JavaScript to design a simple calculator to perform the following operations: sum, product, difference and quotient.<br>10. Write a simple program in PHP by installing and configuring XAMPP with editor (Sublime, Dream viewer).<br>11. Write a PHP program to store current date-time in a COOKIE and display the 'Last visited on' date-time on the web page upon reopening of the same page<br>12. Write a PHP program to store page views count in SESSION, to increment the count on each refresh, and to show the count on web page.<br>13. Write a PHP program to understand file handling.<br>14. Program on Arrays & Functions in PHP<br>15. PHP program on Superglobals.<br>16. . Program to understand the difference between GET and POST method | In<br><br>accordance with theory of BTCS 512 |

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| | 17. Write a PHP program to display a digital clock which displays the current time of the server.<br>18. Write a PHP program on PHP XML Parsers.<br>19. Write a PHP code to make database connection, Create Data Base, Create Table in MySQL, delete item from table and close connection.<br>20. Install a database(MySQL or Oracle). Create a table which should contain at least the following fields: name, password, email-id, phone number(these should hold the data from the registration form). | |

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| | **Text Books** |
| 1. | HTML and CSS: Design and Build Websites – Jon Duckett |
| 2. | JavaScript and JQuery: Interactive Front–End Web Development – Jon Duckett |
| | **References** |
| 1. | Greenlaw R and Hepp E "Fundamentals of Internet and www". |
| 2. | B. Underdahle and K.Underdahle, "Internet and Web Page / Website Design", DG Books India (P) Ltd. |
| 3. | Steven Holzner, PHP: The Complete Reference |
| 4. | Robin Nixon, Learning PHP, MYSQL, JavaScript, CSS & HTML5, O'REILLY Media, Inc. |